

Solve missing-number expressions

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Solve $6 \times (n + 5) = 72$. Copy or complete the first step.

2. Solve $180 \div (n - 2) = 30$.

3. Find n : $14 + n \times 5 = 64$.

4. Miro says: Miro divides 72 by 6 but forgets to subtract the 5 inside the brackets. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Solve $6 \times (n + 5) = 72$. Copy or complete the first step.
Answer $n = 7$.
2. Solve $180 \div (n - 2) = 30$.
Answer $n = 8$.
3. Find n : $14 + n \times 5 = 64$.
Answer $n = 10$.
4. Miro says: Miro divides 72 by 6 but forgets to subtract the 5 inside the brackets. Is this correct? Explain.
Answer Undo the outer multiplication first, then undo the addition inside the brackets.

Solve missing-number expressions

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Solve $6 \times (n + 5) = 72$.

6. Check this result using an inverse, estimate or known fact:
Find n : $14 + n \times 5 = 64$.

2. Solve $180 / (n - 2) = 30$.

7. Create two different expressions using $n = 4$ that both equal 36.

3. Find n : $14 + n \times 5 = 64$.

8. Identify and correct this misconception: Miro divides 72 by 6 but forgets to subtract the 5 inside the brackets.

4. Solve this independently and show every stage: Solve $6 \times (n + 5) = 72$.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Solve $180 / (n - 2) = 30$.

10. Write and solve a similar question to: Find n : $14 + n \times 5 = 64$.

Teacher answers and checking prompts

1. Solve $6 \times (n + 5) = 72$.
Answer $n = 7$.
2. Solve $180 \div (n - 2) = 30$.
Answer $n = 8$.
3. Find n : $14 + n \times 5 = 64$.
Answer $n = 10$.
4. Solve this independently and show every stage: Solve $6 \times (n + 5) = 72$.
Answer $n = 7$.
5. Use a second method or representation for: Solve $180 \div (n - 2) = 30$.
Answer Expected result: $n = 8$. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Find n : $14 + n \times 5 = 64$.
Answer Expected result: $n = 10$. A valid check must be shown.
7. Create two different expressions using $n = 4$ that both equal 36.
Answer Examples include $5 \times (4 + 3) + 1$ and $40 - 4$.
8. Identify and correct this misconception: Miro divides 72 by 6 but forgets to subtract the 5 inside the brackets.
Answer Undo the outer multiplication first, then undo the addition inside the brackets.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Find n : $14 + n \times 5 = 64$.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Solve missing-number expressions

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Create two different expressions using $n = 4$ that both equal 36.

6. Create a new example that tests the same idea as: Find n : $14 + n \times 5 = 64$.

2. Prove the answer to this question, rather than only calculating it: Solve $6 \times (n + 5) = 72$.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Solve $180 / (n - 2) = 30$.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: $n = 10$.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro divides 72 by 6 but forgets to subtract the 5 inside the brackets.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Create two different expressions using $n = 4$ that both equal 36.
Answer Examples include $5 \times (4 + 3) + 1$ and $40 - 4$.
2. Prove the answer to this question, rather than only calculating it: Solve $6 \times (n + 5) = 72$.
Answer Expected result: $n = 7$. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Solve $180 / (n - 2) = 30$.
Answer Expected result: $n = 8$. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: $n = 10$.
Answer One valid reconstruction is: Find n : $14 + n \times 5 = 64$. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro divides 72 by 6 but forgets to subtract the 5 inside the brackets.
Answer Undo the outer multiplication first, then undo the addition inside the brackets.
6. Create a new example that tests the same idea as: Find n : $14 + n \times 5 = 64$.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.